MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Outotec
LARSEN & TOUBRO

Project: **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **CALCULATION**

49PR POWER PLANT COMPRESSOR ROOM

DESIGN CALCULATION OF FOUNDATION

AS BUILT

MD-216-49PR-EG-CV-CAL-2012

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
D01	27-10-2023	AS BUILT	SHD	SGC	PRI	
C01	29-08-2020	ISSUED FOR CONSTRUCTION	SHD	SGC	PRI	
A01	31-07-2020	FIRST ISSUE	SHD	SGC	PRI	

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Ma'aden Document Number:

MD-216-49PR-EG-CV-CAL-2012 D01

Outotec Document Number:

900707 GAU01 WG01 CED01 45020 OU100688145 2

L&T Document Number:

O19045 008 C K 24293 004 2

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Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2	3	
Date	31-07-2020	29-08-2020	27-10-2023				Date	31-07-2020	29-08-2020	27-10-2023		
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Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

REFERENCE DOCUMENTS

DOCUMENT NUMBER	DOCUMENT DESCRIPTION
MD-216-1000-EG-ST-CRT-1001	Civil Structural Design Criteria
MD-216-3000-EG-CV-RPT-1008	Geotechnical Report
SBC 301 :2007	Saudi Structural - Loading & Forces Code
SBC 304 :2018	Saudi Concrete Structures Code
ACI 318 :2014	Building Code Requirement of Structural Concrete

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Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document
LRFD	Load and Resistance Factor Design
F.O.S	Factor of Safety

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1. INTENT

The document is prepared for Design Calculation of Column Foundation for **COMPRESSOR ROOM**

2. SCOPE

The document is prepared for Design Calculation of Column Foundation for COMPRESSOR ROOM
All the loads applied to the structure for analysis & design followed Civil -Structural Design Criteria and SBC 301:2007
Design of Reinforced Concrete Foundations followed Strength Design as per SBC 304:2018 & ACI 318:2008
Net Safe Bearing Capaciity is obtained from Geotechnical report BH28 . Approximate value of 50T/sq.m is considered

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3 Design of Compressor Room Column Foundation

3.1 Design of Foundation mkd. F1

(Refer Foundation Detail dwg no. MD-216-49PR-EG-CV-GAD-2013 & Foundation Load Data)

Load Detail	Col Mkd. A5		
	FX (T)	FY (T)	FZ (T)
1 DL	-0.2	5.6	0.1
2 LLROOF	-0.1	0.9	0.0
3 MR	0.0	1.5	0.0
21 WL1(+X)S	1.1	-1.3	0.1
22 WL2(+X)P	0.9	-1.6	0.2
23 WL3(-X)S	-0.6	0.8	0.3
24 WL4(-X)P	-0.9	0.6	0.5
25 WL5(+Z)S	-0.2	0.3	0.4
26 WL6(+Z)P	-0.4	0.0	0.6
27 WL7(-Z)S	-1.0	-1.0	-0.8
28 WL8(-Z)P	-0.4	-1.1	-0.4

Strength Load Combinations as per Cl. No. 2.3.2, SBC 301:2007 for the most critical support reactions as follows:

STRENGTH DESIGN	Col Mkd. A5		
Load Combinations	FX (T)	FY (T)	FZ (T)
1.4D	-0.25	7.84	0.12
1.4D+1.7(MR)+0.5Lr	-0.3	10.8	0.1
1.2D+1.6Lr+1.0(MR)	-0.4	9.6	0.1
1.2D + 1.6WL1+0.5Lr	1.5	5.1	0.2
1.2D + 1.6WL2+0.5Lr	1.1	4.6	0.4
1.2D + 1.6WL3+0.5Lr	-1.3	8.4	0.5
1.2D + 1.6WL4+0.5Lr	-1.7	8.1	0.8
1.2D + 1.6WL5+0.5Lr	-0.6	7.7	0.8
1.2D + 1.6WL6+0.5Lr	-1.0	7.2	1.0
1.2D + 1.6WL7+0.5Lr	-1.9	5.5	-1.1
1.2D + 1.6WL8+0.5Lr	-0.9	5.3	-0.6
0.9D+1.6WL1	1.7	3.0	0.2
0.9D+1.6WL2	1.2	2.5	0.4
0.9D+1.6WL3	-1.1	6.3	0.5
0.9D+1.6WL4	-1.5	6.0	0.8
0.9D+1.6WL5	-0.5	5.6	0.7
0.9D+1.6WL6	-0.8	5.1	1.0
0.9D+1.6WL7	-1.7	3.4	-1.1
0.9D+1.6WL8	-0.8	3.2	-0.6

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DESIGN OF FOUNDATION MKD. F1

	ABOUT-X	ABOUT-Z
Size of column :	0.55m	0.55m
Size of fdn. raft :	1.65m	1.65m
Thickness of raft :	0.30 m	
Original ground level:	0.00 m	
Finished floor level :	0.00 m	
U/S level of foundation :	-1.50 m	
U/S level Base Plate :	0.50 m	
Unit weight of conc :	2.5 T/m^3	
Unit weight of soil :	1.8 T/m^3	

SYMBOLS AND NOTATIONS

Dead Load	D
live load	LL
Roof live load	Lr
Wind Load	W
Monorail Load	MR

Design Input:

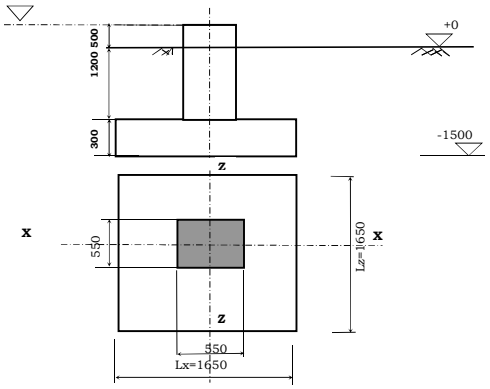
Characteristic strength of concrete, f '$_c$=	35 MPa
Yield stress of steel reinforcement, f_y=	420 MPa
Clear cover for main reinforcement bars =	75mm
Dia. of bars provd. along X direction at bottom =	16mm
Dia. of bars provd. along X direction at top =	12mm
Dia. of bars provided along Z direction at bottom=	16mm
Dia. of bars provided along Z direction at top=	12mm
Mininum percentage of reinforcement at bottom=	0.12%
Mininum percentage of reinforcement at top=	0.06%

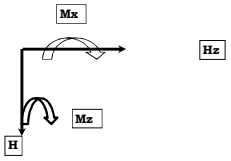
Calculation of Weight of Foundation :

Weight of raft & Col.= (1.65*1.65*0.3*2.5)+(0.55*0.55*1.7*2.5) = 3.33T

Weight of soil = ((1.65*1.65)-(0.55*0.55))*1.2*1.8 = 5.23T

U/S Base Pl Total = 8.55T





Effective depth of slab along X direction = 217mm = 0.217m

Effective depth of slab along Z direction = 201mm = 0.201m

Clear projection along Z direction = 0.55m

Clear projection along X direction = 0.55m

WORKING STRESS LOAD COMBINATION (SBC 301-2.4.1)

WORKING STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation				
Load Case	FX (T)	FY(T)	FZ(T)	M_X	M_Z	V (T)	Hx (T)	MX (T-m)	Hz (T)	MZ (T-m)
DL+Lr	-0.33	6.45	0.09	0.0	0.00	15.00	-0.33	0.18	0.09	0.65
DL+MR	-0.18	7.10	0.09	0.0	0.00	15.65	-0.18	0.18	0.09	0.36
DL+0.75*MR+0.75*Lr	-0.29	7.37	0.09	0.0	0.00	15.92	-0.29	0.18	0.09	0.58
DL+0.75*Lr+0.75*WL1	0.56	5.28	0.13	0.0	0.00	13.83	0.56	0.27	0.13	1.12
DL+0.75*Lr+0.75*WL2	0.36	5.05	0.23	0.0	0.00	13.60	0.36	0.46	0.23	0.71
DL+0.75*Lr+0.75*WL3	-0.74	6.83	0.28	0.0	0.00	15.38	-0.74	0.56	0.28	1.48
DL+0.75*Lr+0.75*WL4	-0.93	6.67	0.43	0.0	0.00	15.22	-0.93	0.86	0.43	1.86
DL+0.75*Lr+0.75*WL5	-0.43	6.48	0.40	0.0	0.00	15.03	-0.43	0.80	0.40	0.85
DL+0.75*Lr+0.75*WL6	-0.61	6.26	0.50	0.0	0.00	14.81	-0.61	1.01	0.50	1.22
DL+0.75*Lr+0.75*WL7	-1.02	5.49	-0.48	0.0	0.00	14.04	-1.02	0.97	-0.48	2.04
DL+0.75*Lr+0.75*WL8	-0.59	5.40	-0.22	0.0	0.00	13.95	-0.59	0.44	-0.22	1.18
0.6*DL+WL1	1.02	2.09	0.11	0.0	0.00	10.64	1.02	0.23	0.11	2.05
0.6*DL+WL2	0.75	1.77	0.24	0.0	0.00	10.32	0.75	0.49	0.24	1.50
0.6*DL+WL3	-0.71	4.15	0.31	0.0	0.00	12.70	-0.71	0.62	0.31	1.42
0.6*DL+WL4	-0.96	3.93	0.51	0.0	0.00	12.48	-0.96	1.01	0.51	1.92
0.6*DL+WL5	-0.29	3.68	0.47	0.0	0.00	12.23	-0.29	0.93	0.47	0.58
0.6*DL+WL6	-0.53	3.39	0.61	0.0	0.00	11.94	-0.53	1.21	0.61	1.07
0.6*DL+WL7	-1.09	2.36	-0.71	0.0	0.00	10.91	-1.09	1.42	-0.71	2.17
0.6*DL+WL8	-0.51	2.23	-0.36	0.0	0.00	10.78	-0.51	0.71	-0.36	1.02

CHECK FOR OVERTURNING AND SLIDING

	Load at Foundation Base					Sliding Coeffecient, μ = 0.4			
	V (T)	Hx (T)	Mz (Tm)	Hz (T)	Mx (Tm)	F.O.S. Mz	F.O.S. Mx	F.O.S. Hx	F.O.S. Hz
Load Case									
DL+Lr	15.00	-0.33	0.65	0.09	0.18	18.99	70.33	18.41	68.20
DL+MR	15.65	-0.18	0.36	0.09	0.18	35.48	73.37	34.40	71.15
DL+0.75*MR+0.75*Lr	15.92	-0.29	0.58	0.09	0.18	22.64	74.61	21.95	72.35
DL+0.75*Lr+0.75*WL1	13.83	0.56	1.12	0.13	0.27	10.18	42.43	9.87	41.14
DL+0.75*Lr+0.75*WL2	13.60	0.36	0.71	0.23	0.46	15.80	24.18	15.33	23.45
DL+0.75*Lr+0.75*WL3	15.38	-0.74	1.48	0.28	0.56	8.57	22.66	8.31	21.97
DL+0.75*Lr+0.75*WL4	15.22	-0.93	1.86	0.43	0.86	6.76	14.65	6.55	14.21
DL+0.75*Lr+0.75*WL5	15.03	-0.43	0.85	0.40	0.80	14.54	15.56	14.10	15.09
DL+0.75*Lr+0.75*WL6	14.81	-0.61	1.22	0.50	1.01	10.05	12.13	9.74	11.77
DL+0.75*Lr+0.75*WL7	14.04	-1.02	2.04	-0.48	0.97	5.67	11.98	5.50	11.62
DL+0.75*Lr+0.75*WL8	13.95	-0.59	1.18	-0.22	0.44	9.75	26.21	9.45	25.41
0.6*DL+WL1	10.64	1.02	2.05	0.11	0.23	4.28	38.21	4.15	37.06
0.6*DL+WL2	10.32	0.75	1.50	0.24	0.49	5.67	17.40	5.50	16.87
0.6*DL+WL3	12.70	-0.71	1.42	0.31	0.62	7.39	16.96	7.16	16.45
0.6*DL+WL4	12.48	-0.96	1.92	0.51	1.01	5.36	10.16	5.20	9.85
0.6*DL+WL5	12.23	-0.29	0.58	0.47	0.93	17.33	10.81	16.80	10.48
0.6*DL+WL6	11.94	-0.53	1.07	0.61	1.21	9.24	8.12	8.96	7.87
0.6*DL+WL7	10.91	-1.09	2.17	-0.71	1.42	4.15	6.35	4.02	6.15
0.6*DL+WL8	10.78	-0.51	1.02	-0.36	0.71	8.74	12.45	8.47	12.08

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ULTIMATE STRESS DESIGN	Load at U/S Of Base Plt					Load at bottom of foundation		
Load Case	FX (T)	FY(T)	FZ(T)	M_x	M_z	V (T)	Mx (T-m)	Mz (T-m)
1.4D	-0.25	7.84	0.12	0.0	0.00	16.394	0.25	0.51
1.4D+1.7(MR)+0.5Lr	-0.33	10.82	0.12	0.0	0.00	19.370	0.25	0.65
1.2D+1.6Lr+1.0(MR)	-0.45	9.59	0.11	0.0	0.00	18.136	0.21	0.90
1.2D + 1.6WL1+0.5Lr	1.52	5.11	0.20	0.0	0.00	13.657	0.41	3.05
1.2D + 1.6WL2+0.5Lr	1.09	4.61	0.41	0.0	0.00	13.160	0.83	2.17
1.2D + 1.6WL3+0.5Lr	-1.25	8.41	0.52	0.0	0.00	16.957	1.03	2.50
1.2D + 1.6WL4+0.5Lr	-1.65	8.07	0.83	0.0	0.00	16.616	1.66	3.31
1.2D + 1.6WL5+0.5Lr	-0.58	7.66	0.77	0.0	0.00	16.210	1.54	1.16
1.2D + 1.6WL6+0.5Lr	-0.97	7.19	0.99	0.0	0.00	15.742	1.98	1.94
1.2D + 1.6WL7+0.5Lr	-1.85	5.55	-1.11	0.0	0.00	14.098	2.23	3.70
1.2D + 1.6WL8+0.5Lr	-0.93	5.34	-0.55	0.0	0.00	13.894	1.10	1.86
0.9D+1.6WL1	1.65	3.00	0.18	0.0	0.00	11.550	0.36	3.30
0.9D+1.6WL2	1.21	2.50	0.39	0.0	0.00	11.053	0.77	2.42
0.9D+1.6WL3	-1.12	6.30	0.49	0.0	0.00	14.850	0.98	2.25
0.9D+1.6WL4	-1.53	5.96	0.81	0.0	0.00	14.509	1.61	3.05
0.9D+1.6WL5	-0.46	5.55	0.74	0.0	0.00	14.103	1.48	0.91
0.9D+1.6WL6	-0.84	5.09	0.97	0.0	0.00	13.635	1.93	1.68
0.9D+1.6WL7	-1.73	3.44	-1.14	0.0	0.00	11.991	2.28	3.45
0.9D+1.6WL8	-0.80	3.24	-0.58	0.0	0.00	11.787	1.15	1.61

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Check for Bearing Pressure

Load Combination & Calculation of Base pressure

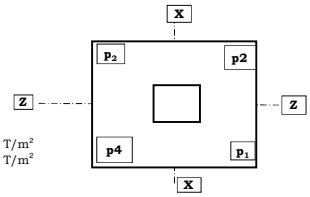
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Appox. Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² (BH28) For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	Base Pressure (T/m²) V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)	Remarks
DL+Lr	15.00	0.65	0.18	5.51	0.87	0.24	8.62	8.15	6.88	6.41	5.92	O.K
DL+MR	15.65	0.36	0.18	5.75	0.49	0.24	8.47	8.00	7.50	7.03	5.77	O.K
DL+0.75*MR+0.75*Lr	15.92	0.58	0.18	5.85	0.77	0.24	8.86	8.39	7.31	6.84	6.16	O.K
DL+0.75*Lr+0.75*WL1	13.83	1.12	0.27	5.08	1.50	0.36	8.94	8.22	5.94	5.22	6.24	O.K
DL+0.75*Lr+0.75*WL2	13.60	0.71	0.46	5.00	0.95	0.62	8.56	7.32	6.67	5.43	5.86	O.K
DL+0.75*Lr+0.75*WL3	15.38	1.48	0.56	5.65	1.98	0.75	10.37	8.88	6.42	4.92	7.67	O.K
DL+0.75*Lr+0.75*WL4	15.22	1.86	0.86	5.59	2.48	1.14	11.22	8.93	6.25	3.96	8.52	O.K
DL+0.75*Lr+0.75*WL5	15.03	0.85	0.80	5.52	1.14	1.06	9.72	7.60	7.45	5.32	7.02	O.K
DL+0.75*Lr+0.75*WL6	14.81	1.22	1.01	5.44	1.62	1.35	10.41	7.72	7.16	4.47	7.71	O.K
DL+0.75*Lr+0.75*WL7	14.04	2.04	0.97	5.16	2.73	1.29	11.18	8.60	5.72	3.14	8.48	O.K
DL+0.75*Lr+0.75*WL8	13.95	1.18	0.44	5.12	1.58	0.59	9.28	8.11	6.13	4.96	6.58	O.K
0.6*DL+WL1	10.64	2.05	0.23	3.91	2.74	0.31	8.95	8.34	3.48	2.86	6.25	O.K
0.6*DL+WL2	10.32	1.50	0.49	3.79	2.01	0.65	8.45	7.14	4.44	3.13	5.75	O.K
0.6*DL+WL3	12.70	1.42	0.62	4.66	1.89	0.82	9.38	7.73	5.59	3.94	6.68	O.K
0.6*DL+WL4	12.48	1.92	1.01	4.59	2.57	1.35	10.51	7.80	5.37	2.66	7.81	O.K
0.6*DL+WL5	12.23	0.58	0.93	4.49	0.78	1.25	8.52	6.02	6.96	4.47	5.82	O.K
0.6*DL+WL6	11.94	1.07	1.21	4.39	1.42	1.62	9.43	6.19	6.58	3.34	6.73	O.K
0.6*DL+WL7	10.91	2.17	1.42	4.01	2.90	1.89	10.80	7.01	5.00	1.21	8.10	O.K
0.6*DL+WL8	10.78	1.02	0.71	3.96	1.36	0.95	8.28	6.37	5.55	3.65	5.58	O.K

Calculation of Design Pressure

Load Combination & Calculation of Base pressure

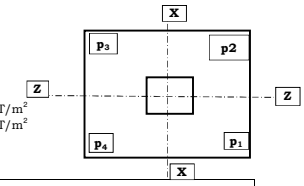
Max. Base pressure = $p_{max} = (V/(Lx.Lz)) + (6.Mz/((Lx)^2.Lz) + (6.Mx/(Lx.(Lz)^2)$

Min. Base pressure = $p_{min} = (V/(Lx.Lz)) - (6.Mz/((Lx)^2.Lz) - (6.Mx/(Lx.(Lz)^2)$

Surcharge load = 2 T/m²

Net SBC of soil for DL, LL= 50 T/m² For WL/SL case = 62.5 T/m²

Gross SBC of soil for DL, LL= 53.6 T/m² For WL/SL case = 66.1 T/m²



Load Case	Design Load/moments V (T)	Mz (T-m)	Mx (T-m)	V/ (Lx.Lz)	6.Mz/ (Lx)².Lz	6.Mx/ Lx.(Lz)²	p1 (pmax)	p2	p4	p3 (pmin)	(pmax-γh)
1.4D	16.4	0.51	0.25	6.02	0.68	0.33	9.03	8.37	7.67	7.01	6.33
1.4D+1.7(MR)+0.5Lr	19.4	0.65	0.25	7.11	0.87	0.33	10.32	7.66	6.57	5.91	8.16
1.2D+1.6Lr+1.0(MR)	18.1	0.90	0.21	6.66	1.20	0.28	10.14	7.58	5.74	5.18	7.98
1.2D + 1.6WL1+0.5Lr	13.7	3.05	0.41	5.02	4.07	0.55	11.63	8.54	1.49	0.40	9.47
1.2D + 1.6WL2+0.5Lr	13.2	2.17	0.83	4.83	2.90	1.10	10.84	6.63	3.04	0.83	8.68
1.2D + 1.6WL3+0.5Lr	17.0	2.50	1.03	6.23	3.34	1.38	12.94	8.19	4.26	1.51	10.78
1.2D + 1.6WL4+0.5Lr	16.6	3.31	1.66	6.10	4.42	2.22	14.74	8.30	3.91	-0.54	12.58
1.2D + 1.6WL5+0.5Lr	16.2	1.16	1.54	5.95	1.55	2.05	11.56	5.46	6.45	2.35	9.40
1.2D + 1.6WL6+0.5Lr	15.7	1.94	1.98	5.78	2.59	2.65	13.02	5.72	5.84	0.54	10.86
1.2D + 1.6WL7+0.5Lr	14.1	3.70	2.23	5.18	4.95	2.97	15.10	7.15	3.21	-2.74	12.94
1.2D + 1.6WL8+0.5Lr	13.9	1.86	1.10	5.10	2.49	1.47	11.06	6.12	4.09	1.15	8.90
0.9D+1.6WL1	11.5	3.30	0.36	4.24	4.41	0.48	11.13	8.18	0.31	-0.64	8.97
0.9D+1.6WL2	11.1	2.42	0.77	4.06	3.24	1.03	10.33	6.27	1.85	-0.21	8.17
0.9D+1.6WL3	14.8	2.25	0.98	5.45	3.00	1.31	11.76	7.15	3.76	1.15	9.60
0.9D+1.6WL4	14.5	3.05	1.61	5.33	4.08	2.15	13.56	7.26	3.40	-0.90	11.40
0.9D+1.6WL5	14.1	0.91	1.48	5.18	1.22	1.98	10.38	4.41	5.95	1.98	8.22
0.9D+1.6WL6	13.6	1.68	1.93	5.01	2.25	2.58	11.84	4.68	5.34	0.18	9.68
0.9D+1.6WL7	12.0	3.45	2.28	4.40	4.61	3.05	14.06	5.97	2.84	-3.25	11.90
0.9D+1.6WL8	11.8	1.61	1.15	4.33	2.15	1.54	10.02	4.94	3.72	0.64	7.86

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	31-07-2020	29-08-2020	27-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	31-07-2020	29-08-2020	27-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

Calculation of Bending Moment and Shear Force in X direction :

BM_{1x} =Moment at the face of column at section 1x-1x in T-m/m.

BM_{1z} =Moment at the face of column 1z-1z in T-m/m.

SF (One way)= Oneway shear force at a distance 'd' from the face of column in T/m.

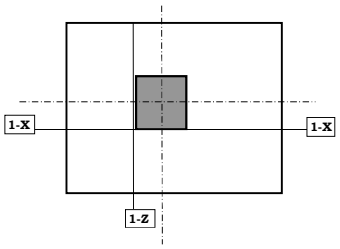
SF (punching) = Punching Shear Force at a distance 0.5d from the face of column in T/m.

Punching perimeter= ((0.55+2*0.5*0.201)+(0.55+2*0.5*0.201))*2

= 3.004 m

Punching Area= ((0.55+2*0.5*0.201)*(0.55+2*0.5*0.201))

= 0.56 m^2 Punching shear stress to be checked



Downward pressure due to self weight of raft and soil = (0.3*2.5+1.2*1.8)

= 2.91 T/m^2

Area of footing, A_f= 2.72 m^2

Check against Two-way shear force at a distance d/2 from face of column:

Design punching shear at a distance d/2 from face of column, Vu= 27.98 T

Strength of Concrete in shear Vc1 = 129.14 T **Table 22.6.5.2_ (a)**

Strength of Concrete in shear Vc2 = 139.21 T **Table 22.6.5.2_ (b)**

Strength of Concrete in shear Vc3 = 119.07 T **Table 22.6.5.2_ (c)**

Design Vc= 119.07 T

Allowable shear stress, (ϕ Vc)/bd = 1.48 N/mm^2

Design punching shear stress, τ = (27.98*10000)/(3004*201)

= 0.46 N/mm^2

No shear reinforcement reqd.

Load Case	p_{1x} (T/m^2)	$B.M_{1x}$ T™	S.F(oneway) (T)	$B.M_{1z}$	S.F(Oneway) (T)	S.F(punching)	V_u/b_0d	Punching check (Remarks)	Allowable shear (0.75*√(fc)/6 (N/mm^2)	Actual Shear	Remarks	Mu/bd2
Ultimate Conditions												
1.4D	6.33	0.96	2.11	0.96	2.21	13.69	0.23	OK	0.74	0.11	OK	0.24
1.4D+1.7(MR)+0.5Lr	8.16	1.23	2.72	1.23	2.85	17.65	0.29	OK	0.74	0.14	OK	0.31
1.2D+1.6Lr+1.0(MR)	7.98	1.21	2.66	1.21	2.79	17.26	0.29	OK	0.74	0.14	OK	0.30
1.2D + 1.6WL1+0.5Lr	9.47	1.43	3.15	1.43	3.31	20.48	0.34	OK	0.74	0.16	OK	0.35
1.2D + 1.6WL2+0.5Lr	8.68	1.31	2.89	1.31	3.03	18.77	0.31	OK	0.74	0.15	OK	0.32
1.2D + 1.6WL3+0.5Lr	10.78	1.63	3.59	1.63	3.76	23.31	0.39	OK	0.74	0.19	OK	0.40
1.2D + 1.6WL4+0.5Lr	12.58	1.90	4.19	1.90	4.39	27.20	0.45	OK	0.74	0.22	OK	0.47
1.2D + 1.6WL5+0.5Lr	9.40	1.42	3.13	1.42	3.28	20.33	0.34	OK	0.74	0.16	OK	0.35
1.2D + 1.6WL6+0.5Lr	10.86	1.64	3.62	1.64	3.79	23.48	0.39	OK	0.74	0.19	OK	0.41
1.2D + 1.6WL7+0.5Lr	12.94	1.96	4.31	1.96	4.52	27.98	0.46	OK	0.74	0.23	OK	0.48
1.2D + 1.6WL8+0.5Lr	8.90	1.35	2.96	1.35	3.11	19.25	0.32	OK	0.74	0.16	OK	0.33
0.9D+1.6WL1	8.97	1.36	2.99	1.36	3.13	19.40	0.32	OK	0.74	0.16	OK	0.34
0.9D+1.6WL2	8.17	1.24	2.72	1.24	2.85	17.67	0.29	OK	0.74	0.14	OK	0.31
0.9D+1.6WL3	9.60	1.45	3.20	1.45	3.35	20.76	0.34	OK	0.74	0.17	OK	0.36
0.9D+1.6WL4	11.40	1.72	3.80	1.72	3.98	24.65	0.41	OK	0.74	0.20	OK	0.43
0.9D+1.6WL5	8.22	1.24	2.74	1.24	2.87	17.78	0.29	OK	0.74	0.14	OK	0.31
0.9D+1.6WL6	9.68	1.46	3.22	1.46	3.38	20.93	0.35	OK	0.74	0.17	OK	0.36
0.9D+1.6WL7	11.90	1.80	3.96	1.80	4.15	25.73	0.43	OK	0.74	0.21	OK	0.45
0.9D+1.6WL8	7.86	1.19	2.62	1.19	2.74	17.00	0.28	OK	0.74	0.14	OK	0.29

Load Case	kn (Ø=0.75)	p =	A_{st}
Ultimate Conditions			
1.4D	0.32	0.001	217
1.4D+1.7(MR)+0.5Lr	0.41	0.001	217
1.2D+1.6Lr+1.0(MR)	0.4	0.001	217
1.2D + 1.6WL1+0.5Lr	0.47	0.001	217
1.2D + 1.6WL2+0.5Lr	0.43	0.001	217
1.2D + 1.6WL3+0.5Lr	0.54	0.001	217
1.2D + 1.6WL4+0.5Lr	0.63	0.002	434
1.2D + 1.6WL5+0.5Lr	0.47	0.001	217
1.2D + 1.6WL6+0.5Lr	0.54	0.001	217
1.2D + 1.6WL7+0.5Lr	0.65	0.002	434
1.2D + 1.6WL8+0.5Lr	0.44	0.001	217
0.9D+1.6WL1	0.45	0.001	217
0.9D+1.6WL2	0.41	0.001	217
0.9D+1.6WL3	0.48	0.001	217
0.9D+1.6WL4	0.57	0.001	217
0.9D+1.6WL5	0.41	0.001	217
0.9D+1.6WL6	0.48	0.001	217
0.9D+1.6WL7	0.59	0.001	217
0.9D+1.6WL8	0.39	0.001	217

Location	Mux (T-m)	ϕ Kn = M_{u1}/bd^2 (N/mm^2)	Kn (N/mm^2)	ρ=	Ast (mm^2)	Minm. Ast (mm^2)	Bar Dia	Spacing reqd.(mm)	Maxm spacing (mm)	Spacing provd. (mm)
At bottom	1.96	0.48	0.65	0.002	434.00	390.60	12ϕT	260.59	300	200
At top	0.00	0.00	0.00	0.0018	361.80	361.80	12ϕT	312.60	300	200

SUMMARY OF REINFORCEMENT IN RAFT:

Along X Direction :

At Bottom **12ϕT** **200** c/c At Top **12ϕT** **200** c/c

Along Z Direction :

At Bottom **12ϕT** **200** c/c At Top **12ϕT** **200** c/c

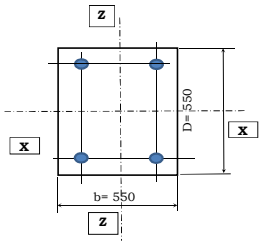
Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01	Outotec / L&T Revision No.:	0	1	2	3
Date	31-07-2020	29-08-2020	27-10-2023	Date	31-07-2020	29-08-2020	27-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	Approved	PRI	PRI	PRI	

DESIGN OF COLUMN MKD P1

Input:	
Clear cover to reinf.=	75mm
Dia. of main reinf.=	25mm
Dia. of link=	10mm
Effective cover =	97.5mm
Grade of Concrete (f_c') =	M35
Grade of Steel (f_y) =	Fy420
Grade of Reinf(links).=	Fy420
Min. Steel Ratio	0.50%
Max. Steel Ratio	4.00%
Pedestal Height	1700

Table 20.6.1.3.1



For Compression-controlled sections
Strength reduction factor ϕ = 0.65 Table 21.2.2

Unsupported length of column about X (l_{ux}) = 1700
Unsupported length of column about Z (l_{uz}) = 1700

For Columns not braced against sidesway,
Slenderness ratio, $K\,l_{ux} / r_x$ = 21.42 < 22 SECTION 6.2.5a
$K\,l_{uz} / r_z$ = 21.42 < 22 SECTION 6.2.5a

Where, Effective length factor K= 2
Radius of gyration r_x= 159
Radius of gyration r_z= 159

γ_y = 0.68
γ_z = 0.68

Load Combination	Factored Load/Moment (T)			Nominal Load/Moment			Kn = Pn/f_c'bD (N/mm^2)
	Hx	V	Hz	Mz (Tm)	V (T)	Mx (Tm)	
1.4D	-0.25	7.84	0.12	0.7	12.1	0.3	0.01
1.4D+1.7(MR)+0.5Lr	-0.33	10.82	0.12	0.9	16.6	0.3	0.02
1.2D+1.6Lr+1.0(MR)	-0.45	9.59	0.11	1.2	14.7	0.3	0.01
1.2D + 1.6WL1+0.5Lr	1.52	5.11	0.20	4.0	7.9	0.5	0.01
1.2D + 1.6WL2+0.5Lr	1.09	4.61	0.41	2.8	7.1	1.1	0.01
1.2D + 1.6WL3+0.5Lr	-1.25	8.41	0.52	3.3	12.9	1.3	0.01
1.2D + 1.6WL4+0.5Lr	-1.65	8.07	0.83	4.3	12.4	2.2	0.01
1.2D + 1.6WL5+0.5Lr	-0.58	7.66	0.77	1.5	11.8	2.0	0.01
1.2D + 1.6WL6+0.5Lr	-0.97	7.19	0.99	2.5	11.1	2.6	0.01
1.2D + 1.6WL7+0.5Lr	-1.85	5.55	-1.11	4.8	8.5	2.9	0.01
1.2D + 1.6WL8+0.5Lr	-0.93	5.34	-0.55	2.4	8.2	1.4	0.01
0.9D+1.6WL1	1.65	3.00	0.18	4.3	4.6	0.5	0.00
0.9D+1.6WL2	1.21	2.50	0.39	3.2	3.9	1.0	0.00
0.9D+1.6WL3	-1.12	6.30	0.49	2.9	9.7	1.3	0.01
0.9D+1.6WL4	-1.53	5.96	0.81	4.0	9.2	2.1	0.01
0.9D+1.6WL5	-0.46	5.55	0.74	1.2	8.5	1.9	0.01
0.9D+1.6WL6	-0.84	5.09	0.97	2.2	7.8	2.5	0.01
0.9D+1.6WL7	-1.73	3.44	-1.14	4.5	5.3	3.0	0.01
0.9D+1.6WL8	-0.80	3.24	-0.58	2.1	5.0	1.5	0.00

REINFORCEMENT

Provide,

4-Nos	25 mm Dia	1964.29 Sq mm	(At Corner)		A_{sc} pvd. =	3221 Sq mm
2-Nos	20 mm Dia	628.57 Sq mm	(At middle)			
2-Nos	20 mm Dia	628.57 Sq mm	(At middle)		Total A_{sc} =	1.0 %

β = P_n / P_0 (range is 0.55 to 0.70) Where, P_0 = 0.85 f_c' (A_g - A_{sc}) + f_y A_{sc} = 1025.6538 T SECTION 22.4.2.2

Load Case	From Chart of SP17(M)-R35-420.7		M_{noz}	M_{nox}	β	α_n = log(0.5)/ log(β)	Interaction ratio	Remarks
	R_{nz}=M_{noz} /f_c'b^{2}D	R_{nx}=M_{nox} /f_c'bD^2						
1.4D	0.05	0.05	29.1	29.12	0.55	1.16	0.02	O.K
1.4D+1.7(MR)+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.02	O.K
1.2D+1.6Lr+1.0(MR)	0.05	0.05	29.1	29.12	0.55	1.16	0.03	O.K
1.2D + 1.6WL1+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.11	O.K
1.2D + 1.6WL2+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.09	O.K
1.2D + 1.6WL3+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.11	O.K
1.2D + 1.6WL4+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.16	O.K
1.2D + 1.6WL5+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.08	O.K
1.2D + 1.6WL6+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.12	O.K
1.2D + 1.6WL7+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.19	O.K
1.2D + 1.6WL8+0.5Lr	0.05	0.05	29.1	29.12	0.55	1.16	0.09	O.K
0.9D+1.6WL1	0.05	0.05	29.1	29.12	0.55	1.16	0.12	O.K
0.9D+1.6WL2	0.05	0.05	29.1	29.12	0.55	1.16	0.10	O.K
0.9D+1.6WL3	0.05	0.05	29.1	29.12	0.55	1.16	0.10	O.K
0.9D+1.6WL4	0.05	0.05	29.1	29.12	0.55	1.16	0.15	O.K
0.9D+1.6WL5	0.05	0.05	29.1	29.12	0.55	1.16	0.07	O.K
0.9D+1.6WL6	0.05	0.05	29.1	29.12	0.55	1.16	0.11	O.K
0.9D+1.6WL7	0.05	0.05	29.1	29.12	0.55	1.16	0.19	O.K
0.9D+1.6WL8	0.05	0.05	29.1	29.12	0.55	1.16	0.08	O.K

Links, **10mm dia @ 200 C/C**

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2	3	
Date	31-07-2020	29-08-2020	27-10-2023				Date	31-07-2020	29-08-2020	27-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

REFERENCES

[1] Structural Loading and Forces ; SBC Code 301: 2007
[2] Saudi Concrete Structures Code ; SBC 304 : 2018
[3] Civil,Structural and Architectural Design Criteria ; MD-216-1000-EG-ST-CRT-1001
[4] ACI 318 : 2014 Building Code Requirement of Structural Concrete
[5] Geotechnical Report No. MD-216-3000-EG-CV-RPT-1008
[6] COMPRESSOR ROOM DETAIL DRAWING NO. MD-216-49PR-EG-ST-GAD-2001
[7] COMPRESSOR ROOM STRUCTURAL DOC NO. MD-216-49PR-EG-ST-CAL-2001

Global Project ID	**900707**
Delivery Type	**ENGINEERING PROCUREMENT CONSTRUCTION**
Project Name	**MANSOURAH MASSARAH GOLD PROJECT**
Customer	**MA'ADEN GOLD AND BASE METALS (MA'ADEN)**
Project Country	**KINGDOM OF SAUDI ARABIA**

Ma'aden Revision No:	A01	C01	D01				Outotec / L&T Revision No.:	0	1	2	3	
Date	31-07-2020	29-08-2020	27-10-2023				Date	31-07-2020	29-08-2020	27-10-2023		
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT				Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT		
Prepared By	SHD	SHD	SHD				Prepared By	SHD	SHD	SHD		
Checked	SGC	SGC	SGC				Checked	SGC	SGC	SGC		
Approved	PRI	PRI	PRI				Approved	PRI	PRI	PRI		

ATTACHMENT 1 Foundation Load Data for Compressor Room Building Columns

Global Project ID	900707
Delivery Type	ENGINEERING PROCUREMENT CONSTRUCTION
Project Name	MANSOURAH MASSARAH GOLD PROJECT
Customer	MA'ADEN GOLD AND BASE METALS (MA'ADEN)
Project Country	KINGDOM OF SAUDI ARABIA

Ma'aden Revision No:	A01	C01	D01
Date	31-07-2020	29-08-2020	27-10-2023
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT
Prepared By	SHD	SHD	SHD
Checked	SGC	SGC	SGC
Approved	PRI	PRI	PRI

Outotec / L&T Revision No.:	0	1	2	3
Date	31-07-2020	29-08-2020	27-10-2023	
Revision Text	FIRST ISSUE	ISSUED FOR CONSTRUCTION	AS BUILT	
Prepared By	SHD	SHD	SHD	
Checked	SGC	SGC	SGC	
Approved	PRI	PRI	PRI	

MMGP:POWER PLANT: Compressor Room Load Data

1 DL	Dead load	
2 LLROOF	Live load on roof	
3 MR	Monorail load	
21 WL1(+X)S	Wind load	Wind load in direction of +X Suction(from A grid to B grid)
22 WL2(+X)P	Wind load	Wind load in direction of +X Pressure(from A grid to B grid)
23 WL3(-X)S	Wind load	Wind load in direction of -X Suction(from B grid to A grid)
24 WL4(-X)P	Wind load	Wind load in direction of -X Pressure(from B grid to A grid)
25 WL5(+Z)S	Wind load	Wind load in direction of +Z Suction(from line-1 to line-5)
26 WL6(+Z)P	Wind load	Wind load in direction of +Z Pressure(from line-1 to line-5)
27 WL7(-Z)S	Wind load	Wind load in direction of -Z Suction(from line-5 to line-1)
28 WL8(-Z)P	Wind load	Wind load in direction of -Z Pressure(from line-5 to line-1)

Sign convention

FY	+ve sign indicates vertical load downward
	-ve sign indicates vertical load upward
FX	+ve sign indicates horizonal load in across direction from grid-A to B
	-ve sign indicates horizonal load in across direction from grid-B to A
FZ	+ve sign indicates horizonal load in along direction from line 1 to 5
	-ve sign indicates horizonal load in along direction from line 5 to 1

Monorail and wind load cases are exclusive.

LOAD DATA TABLE

(Refer drg no. MD-216-49PR-EG-ST-GAD-2001 for other details)

	Grid	Load cases	Horizontal Fx Ton	Vertical Fy Ton	Horizontal Fz Ton
line (-1,-2,-3)	B	1 DL	0.2	3	0
		2 LLROOF	0.3	1.7	0
		3 MR	0	1.5	0
		21 WL1(+X)S	0.9	0.5	0
		22 WL2(+X)P	1.5	0.2	0
		23 WL3(-X)S	-1.8	-1.6	0
		24 WL4(-X)P	-1.3	-2	0
		25 WL5(+Z)S	0.4	-0.6	0
		26 WL6(+Z)P	0.8	-1.1	0
		27 WL7(-Z)S	0.4	-0.6	0
		28 WL8(-Z)P	0.8	-1.1	0
line (-1,-2,-3)	A	1 DL	-0.2	3	0
		2 LLROOF	-0.3	1.7	0
		3 MR	0	1.5	0
		21 WL1(+X)S	1.8	-1.6	0
		22 WL2(+X)P	1.3	-2	0
		23 WL3(-X)S	-0.9	0.5	0
		24 WL4(-X)P	-1.4	0.1	0
		25 WL5(+Z)S	-0.4	-0.6	0
		26 WL6(+Z)P	-0.8	-1.1	0
		27 WL7(-Z)S	-0.4	-0.6	0
		28 WL8(-Z)P	-0.8	-1.1	0
line (-4)	B	1 DL	0.2	3.3	-0.1
		2 LLROOF	0.3	1.7	0
		3 MR	0	1.5	0
		21 WL1(+X)S	0.8	0.2	0.1
		22 WL2(+X)P	1.3	-0.4	0.3
		23 WL3(-X)S	-1.6	-1	-0.1
		24 WL4(-X)P	-1.1	-1.4	0
		25 WL5(+Z)S	0.4	-1.3	0.4
		26 WL6(+Z)P	0.8	-1.7	0.4
		27 WL7(-Z)S	0.2	-0.6	-0.3
		28 WL8(-Z)P	0.8	-0.6	-0.3
line (-4)	A	1 DL	-0.2	3.3	-0.1
		2 LLROOF	-0.3	1.7	0
		3 MR	0	1.5	0
		21 WL1(+X)S	1.7	-1.3	0
		22 WL2(+X)P	1.2	-1.5	0
		23 WL3(-X)S	-0.7	0	0.2
		24 WL4(-X)P	-1.2	-0.4	0.3
		25 WL5(+Z)S	-0.4	-1.3	0.4
		26 WL6(+Z)P	-0.8	-1.7	0.4
		27 WL7(-Z)S	-0.5	0.5	-0.6
		28 WL8(-Z)P	-0.8	-0.6	-0.3
line (-5)	B	1 DL	0.2	5.6	0.1
		2 LLROOF	0.1	0.9	0
		3 MR	0	1.5	0
		21 WL1(+X)S	1	0.8	0.2
		22 WL2(+X)P	0.9	0.6	0.5
		23 WL3(-X)S	-1.1	-1.3	0
		24 WL4(-X)P	-0.8	-1.5	0.2
		25 WL5(+Z)S	0.2	0.3	0.4
		26 WL6(+Z)P	0.4	0	0.6
		27 WL7(-Z)S	0	-0.9	-0.5
		28 WL8(-Z)P	0.4	-1.1	-0.4
line-(5)	A	1 DL	-0.2	5.6	0.1
		2 LLROOF	-0.1	0.9	0
		3 MR	0	1.5	0
		21 WL1(+X)S	1.1	-1.3	0.1
		22 WL2(+X)P	0.9	-1.6	0.2
		23 WL3(-X)S	-0.6	0.8	0.3
		24 WL4(-X)P	-0.9	0.6	0.5
		25 WL5(+Z)S	-0.2	0.3	0.4
		26 WL6(+Z)P	-0.4	0	0.6
		27 WL7(-Z)S	-1	-1	-0.8
		28 WL8(-Z)P	-0.4	-1.1	-0.4